

VJ146A2 October 2023 MLR Validation Report

Population-weighted nighttime darkness vs supply-normalized Eskom MLR

2026-06-03

Validation Question

Corrected October 2023 VJ146A2 results show a positive validation relationship between local-overpass Eskom MLR and satellite-observed nighttime darkness.

The validation question is whether days with higher national Eskom manual load reduction at the VIIRS local overpass window also show higher population-weighted darkness in the settlement panel. The headline outcome is the population-weighted share of people in dark settlements, restricted to settlements with population greater than 100.

Methodology

- Date handling: `vj_product_date = date`; `local_overpass_date = VJ product date + 1`.
- Eskom exposure: local-clock hour beginning 01:00 SAST, interpreted as the 01:00-02:00 overpass window.
- Predictor: `shed_share_1_2am_primary = MLR / (MLR + RSA Contracted Demand)`.
- Darkness definitions: strict dark is `p_lit_sett < 0.05`; mostly dark is `p_lit_sett < 0.20`.
- Sample: October 2023 VJ146A2 product dates with matched Eskom local-overpass exposure; settlements with population greater than 100.

The continuous models use all 30 matched October VJ dates. The extreme-bin headline compares the seven highest vs seven lowest shed-share dates. The tied-zero bottom-quartile row is a robustness check because zero-MLR days are tied at the lower quartile.

Headline Table

Table 1: Ravi-style national validation models. Continuous effect is for a 10 pp increase in shed share; bin effects are top minus bottom, in percentage points.

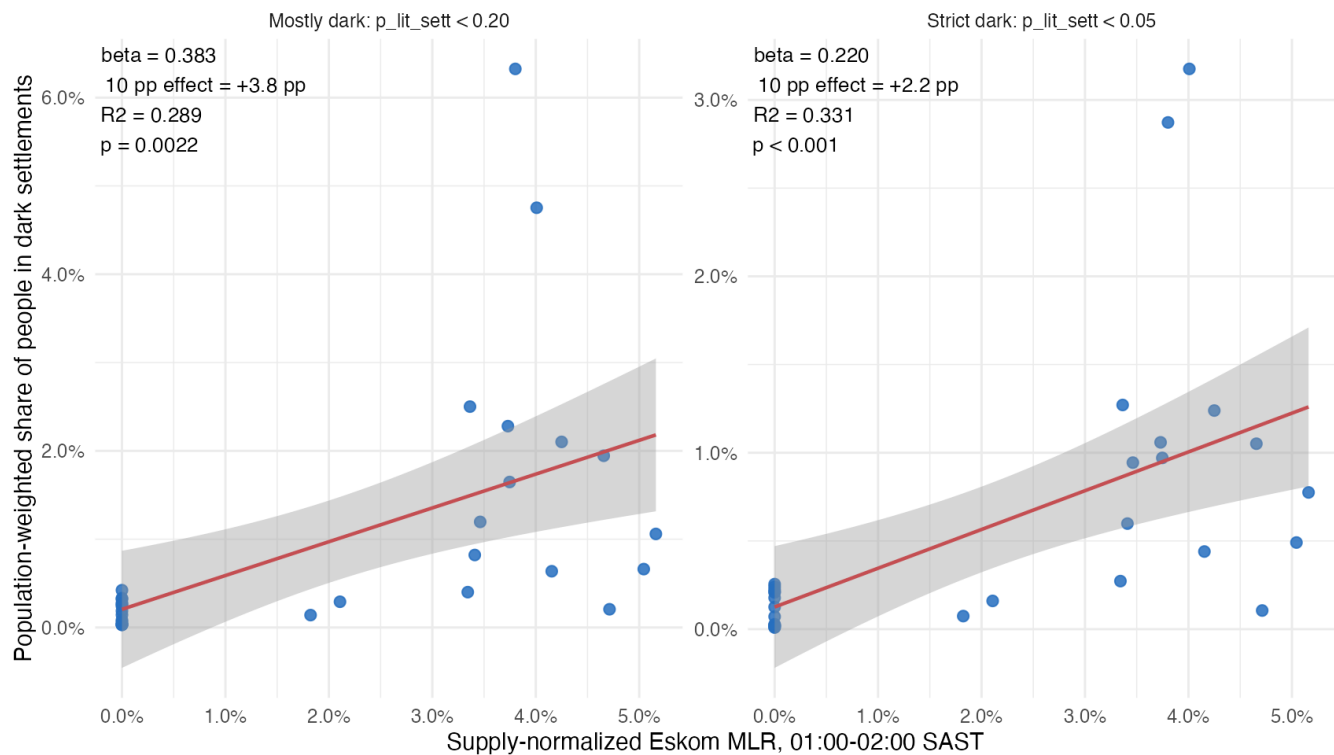
Specification	Outcome	N	Estimate	p	R2	Effect, pp
Headline continuous	Strict (<0.05)	30	0.2197	<0.001	0.331	2.20
Headline continuous	Mostly (<0.20)	30	0.3828	0.0022	0.289	3.83
Headline top 7 vs bottom 7	Strict (<0.05)	14	0.0089	0.0407	0.305	0.89
Headline top 7 vs bottom 7	Mostly (<0.20)	14	0.0142	0.0320	0.329	1.42
Robust tied-zero quartile	Strict (<0.05)	22	0.0115	0.0011	0.422	1.15
Robust tied-zero quartile	Mostly (<0.20)	22	0.0204	0.0021	0.383	2.04

For strict darkness, the continuous coefficient is 0.220 with R2 0.331 and p-value <0.001. A 10 percentage-point increase in shed share implies about 2.2 percentage points higher strict-dark population share.

For mostly-dark settlements, the continuous coefficient is 0.383 with R2 0.289 and p-value 0.0022. A 10 percentage-point increase in shed share implies about 3.8 percentage points higher mostly-dark population share.

Population-weighted darkness vs local-overpass Eskom shed share

Settlements with population > 100; shed share = $\text{MLR} / (\text{MLR} + \text{RSA contracted demand})$

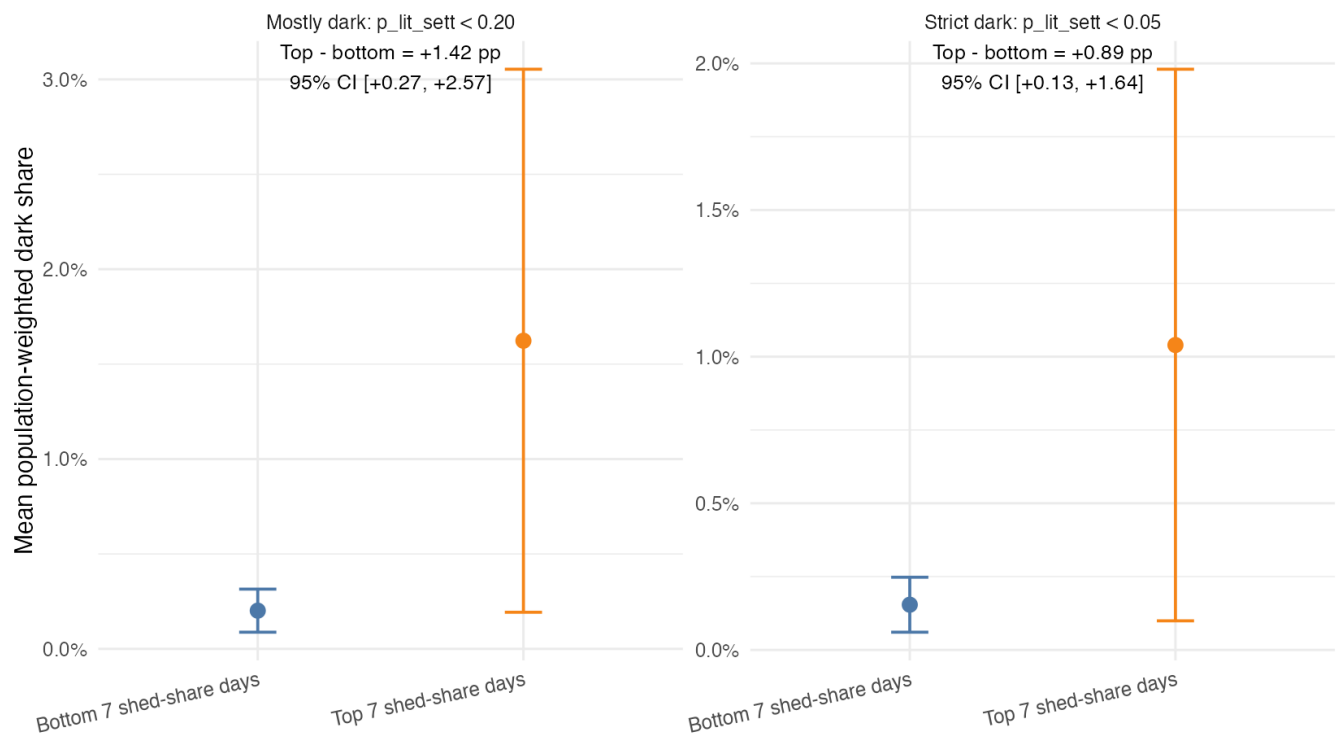


VJ146A2 Oct 2023 only; Eskom date = VJ product date + 1; 01:00-02:00 SAST primary; candidate dark events, not confirmed outages.

Figure 1: Population-weighted darkness increases with supply-normalized Eskom MLR in the corrected local-overpass alignment.

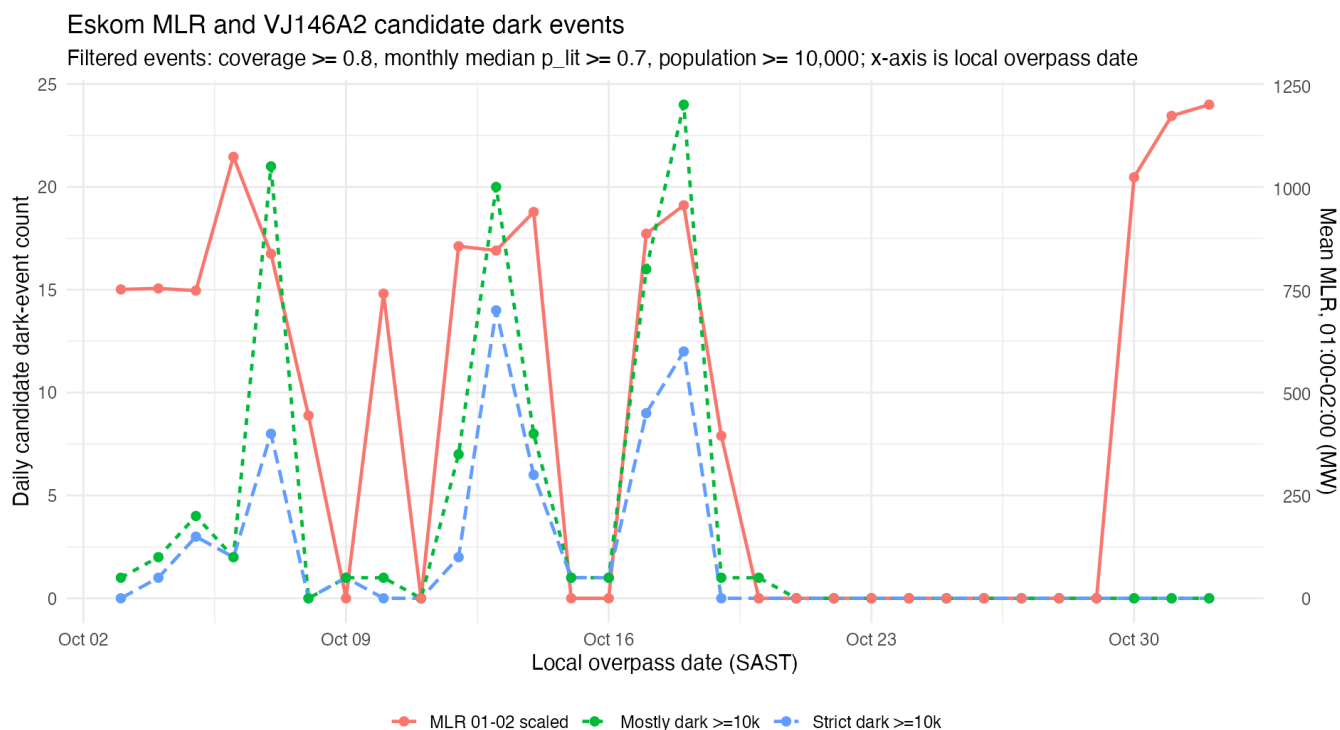
Darkness is higher on top shed-share nights

Rank-based comparison of seven highest vs seven lowest local-overpass shed-share days



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Figure 2: Rank-based top-vs-bottom shed-share comparison. Effects are reported as percentage-point differences in population-weighted darkness.



VJ146A2 Oct 2023 only; Eskom date = VJ product date + 1; 01:00-02:00 SAST primary; candidate dark events, not confirmed outages.

Figure 3: Existing diagnostic time series of Eskom MLR and large candidate dark-event counts.

Figures

Interpretation

The corrected October diagnostic is positive and supervisor-ready as validation evidence: higher local-overpass Eskom MLR is associated with higher satellite-observed darkness. The strongest report-facing specification is the population-weighted national share, because it asks how many people are in settlements that appear dark rather than how many settlement polygons cross a threshold.

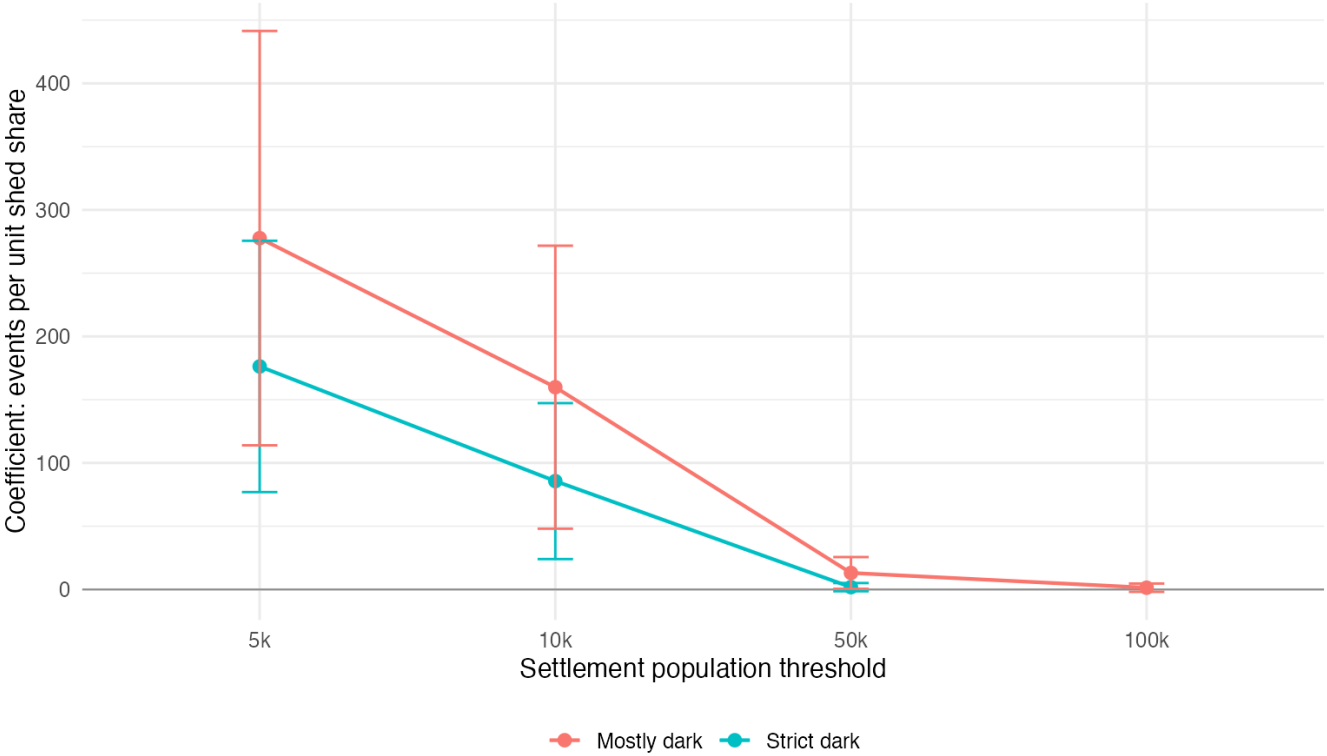
The top-7 vs bottom-7 comparison is a compact robustness view of the same relationship. It should be interpreted as an association around the overpass window, not as proof that individual settlements lost power because of Eskom MLR.

Caveats

- Eskom MLR is a national proxy, while settlement darkness is local.
- Candidate dark events are not confirmed outages.
- VJ146A2 provides one nighttime snapshot, not full-night service continuity.
- Clouds, private generation, local heterogeneity, and acquisition timing can affect settlement-level darkness.
- The date-alignment convention is grounded at the workflow level; it is not per-pixel acquisition-time proof.

Population-threshold sensitivity

OLS coefficient on primary shed share for filtered event counts



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Figure 4: Existing population-threshold sensitivity diagnostic for large candidate event counts.